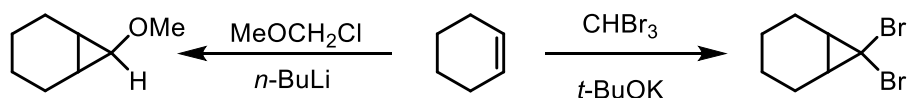
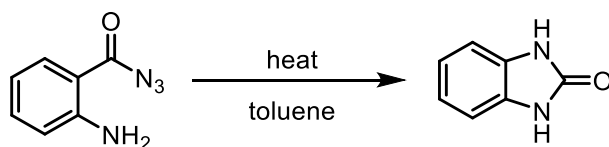


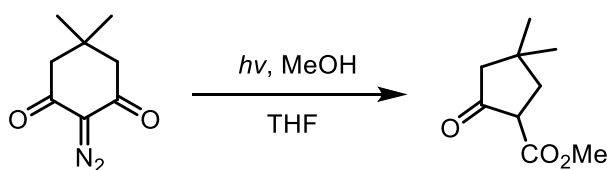
1. Suggest mechanisms for these reactions.



2. Suggest a mechanism for the following transformation.

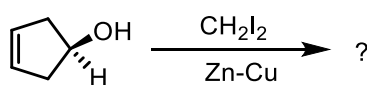


3. Suggest a mechanism for this ring contraction.

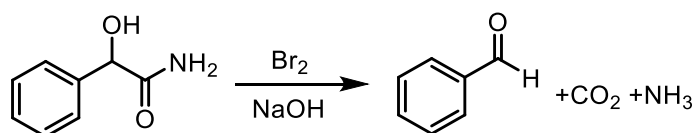


4. Design experiments to figure out whether  $\text{Cl}_2\text{C:}$  is singlet or triplet carbene.

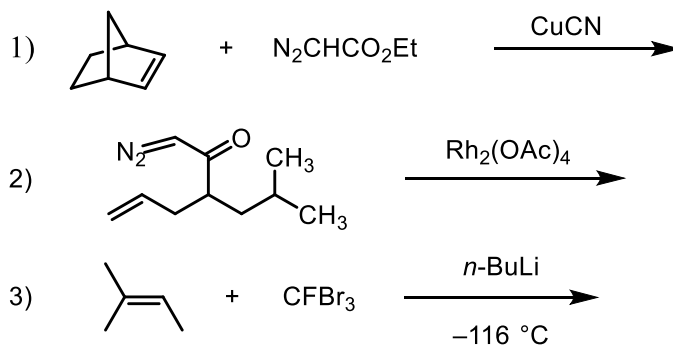
5. Give the major product of this reaction and explain the stereoselectivity.



6. Suggest a mechanism for this reaction.



7. Give the possible products for the following reactions.



8. Use olefin metathesis as the key reaction to synthesize the following compound from starting materials with three or fewer carbons.

